Calculation

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In the mathematical model for the number of fish in the aquarium, the growth factor is now bounded:

$$\frac{dP}{dt} = 0.7\,P\left(1 - \frac{P}{750}\right) - 20,$$

with the initial condition $P(0) = 30$.

In the next exercises, you are going to calculate whether the resulting population stays bounded as well.

Exercise A

2/2 points (graded)

The differential equation has two equilibrium points: P_{e_1} and P_{e_2}. Calculate their values up to one decimal.

P_{e_1}, the smallest of the two:

29.8

Answer: 29.8

29.8

P_{e_2}, the largest of the two:

720.2

Answer: 720.2

720.2

Submit

You have used 1 of 3 attempts

Answers are displayed within the problem

Exercise B

1/1 point (graded)

Sketch (for yourself) the graph of $\frac{dP}{dt}$ as a function of P.

For what value of P does P increase the most?
Give the value with one decimal.

375 ✔ **Answer:** 375.0

375

Submit You have used 2 of 3 attempts

Answers are displayed within the problem

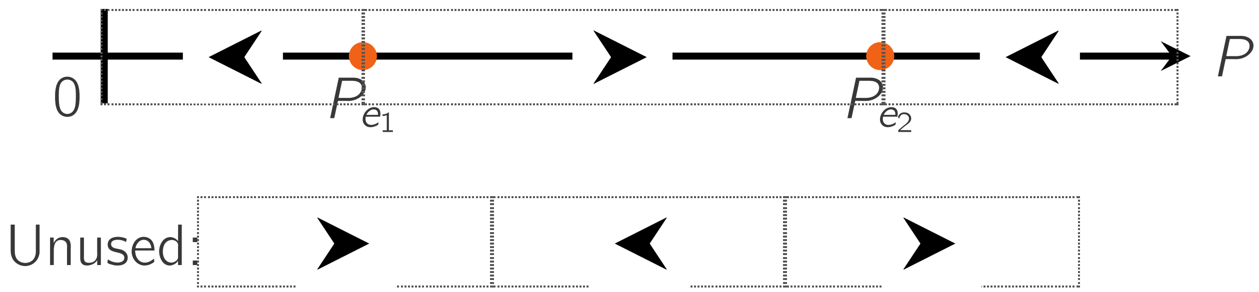
Exercise C

1/1 point (graded)

Keyboard Help

PROBLEM

Using your graph and the information from the previous exercises, construct the phase line.



Submit You have used 2 of 3 attempts.

Reset Show Answer

FEEDBACK

✔ Correctly placed 6 items.

✔ Good work! You have completed the phase line.

✔ Your highest score is 1.0

On the next page, you will validate the solution.